Part 1 - Package Management

```
Ajosh@DESKTOP-SSK364I:~$ mkdir project
Ajosh@DESKTOP-SSK364I:~$ cd project
Ajosh@DESKTOP-SSK364I:~/project$ touch app.txt config.conf
Ajosh@DESKTOP-SSK364I:~/project$ ls
app.txt  config.conf
Ajosh@DESKTOP-SSK364I:~/project$ echo "hello" > app.txt
Ajosh@DESKTOP-SSK364I:~/project$ echo "config" > config.conf
Ajosh@DESKTOP-SSK364I:~/project$ ls
app.txt  config.conf
Ajosh@DESKTOP-SSK364I:~/project$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.6 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [734 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [323 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 1883 kB in 3s (673 kB/s)
24 packages can be upgraded. Run 'apt list --upgradable' to see them.
Ajosh@DESKTOP-SSK364I:~/project$
```

```
Ajosh@DESKTOP-SSK364I:~/project$ sudo apt install tree curl git
[sudo: authenticate] Password:
tree is already the newest version (2.3.1-1).
curl is already the newest version (8.18.0-1ubuntu2.7).
curl set to manually installed.
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 14
Ajosh@DESKTOP-SSK364I:~/project$
```

```
Ajosh@DESKTOP-SSK364I:~/project$ apt search docker
amazon-ecr-credential-helper/resolute 0.7.1-2.1build1 amd64
  Amazon ECR Credential Helper for Docker

auto-apt-proxy/resolute 17.1 all
  automatic detector of common APT proxy settings

buildah/resolute 1.42.1+ds1-2build1 amd64
  CLI tool to facilitate building OCI images

catatonit/resolute 0.2.1-2build1 amd64
  init process for containers

centreon-plugins/resolute 0.0~20250303+dfsg-1 all
  Collection of Nagios plugins to monitor OS, services and network devices

crowdsec/resolute 1.4.6-10.1 amd64
  lightweight and collaborative security engine

crun/resolute 1.21-1ubuntu3 amd64
  lightweight OCI runtime for running containers

cwl-utils/resolute 0.40-1 all
  Utilities for using CWL documents

debcraft/resolute 0.8.1 all
  Easy, fast and secure way to build Debian packages

debocker/resolute 0.2.6build1 all
  docker-powered package builder for Debian

debootstick/resolute 2.8build1 all
  Turn a chroot environment into a bootable image

debuerreotype/resolute 0.16-1 all
  reproducible, snapshot-based Debian rootfs builder

devscripts/resolute 2.26.7 all
  scripts to make the life of a Debian Package maintainer easier

dgx-desktop-docker-gpus/resolute 1.0ubuntu all
lines 1-40...skipping...
amazon-ecr-credential-helper/resolute 0.7.1-2.1build1 amd64
  Amazon ECR Credential Helper for Docker
```

```
Ajosh@DESKTOP-SSK364I:~/project$ apt show nginx
Package: nginx
Version: 1.28.3-2ubuntu1.11
Priority: optional
Section: web
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Debian Nginx Maintainers <pkg-nginx-maintainers@alioth-lists.debian.net>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 1631 kB
Provides: httpd, httpd-cgi, nginx-abi-1.28.3-1
Depends: libc6 (>= 2.34), libcrypt1 (>= 1:4.1.0), libpcre2-8-0 (>= 10.22), libssl3t64 (>= 3.0.0), zlib1g (>= 1:1.1.4), nginx-common (= 1.28.3-2ubuntu1.11)
Breaks: nginx-core (<< 1.22.1-6~), nginx-extras (<< 1.22.1-6~), nginx-light (<< 1.22.1-6~)
Replaces: nginx-core (<< 1.22.1-6~), nginx-extras (<< 1.22.1-6~), nginx-light (<< 1.22.1-6~)
Homepage: https://nginx.org
Download-Size: 617 kB
APT-Manual-Installed: yes
APT-Sources: http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages
Description: small, powerful, scalable web/proxy server
 Nginx ("engine X") is a high-performance web and reverse proxy server
 created by Igor Sysoev. It can be used both as a standalone web server
 and as a proxy to reduce the load on back-end HTTP or mail servers.

Notice: There is 1 additional record. Please use the '-a' switch to see it
Ajosh@DESKTOP-SSK364I:~/project$
```

```
Ajosh@DESKTOP-SSK364I:~/project$ sudo apt remove tree
REMOVING:
  tree

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 14
  Freed space: 125 kB

Continue? [Y/n] y
(Reading database ... 39580 files and directories currently installed.)
Removing tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Ajosh@DESKTOP-SSK364I:~/project$
```

Part 2 - Storage Analysis

```
Ajosh@DESKTOP-SSK364I:~$ cd project
Ajosh@DESKTOP-SSK364I:~/project$ df -h > storage-report.txt
Ajosh@DESKTOP-SSK364I:~/project$ du -sh >> storage-report.txt
Ajosh@DESKTOP-SSK364I:~/project$ lsblk >> storage-report.txt
Ajosh@DESKTOP-SSK364I:~/project$ sudo blkid >> storage-report.txt
[sudo: authenticate] Password:
Ajosh@DESKTOP-SSK364I:~/project$ sudo fdisk -l >> storage-report.txt
Ajosh@DESKTOP-SSK364I:~/project$ cat storage-report.txt
Filesystem      Size  Used Avail Use% Mounted on
none            811M     0  811M   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            811M  4.0K  811M   1% /mnt/wsl
drivers         465G   77G  389G  17% /usr/lib/wsl/drivers
/dev/sdd       1007G  2.0G  954G   1% /
none            811M   44K  811M   1% /mnt/wslg
none            811M     0  811M   0% /usr/lib/wsl/lib
rootfs          806M  2.8M  803M   1% /init
none            811M  544K  811M   1% /run
none            811M     0  811M   0% /run/lock
none            811M     0  811M   0% /run/shm
none            811M   80K  811M   1% /mnt/wslg/versions.txt
none            811M   80K  811M   1% /mnt/wslg/doc
C:\             465G   77G  389G  17% /mnt/c
D:\             310G   83G  228G  27% /mnt/d
E:\             311G  375M  311G   1% /mnt/e
J:\             311G  105M  311G   1% /mnt/j
tmpfs           811M     0  811M   0% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           163M   12K  163M   1% /run/user/1000
8.0K    .
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0     1G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /var/lib/docker
                               /mnt/wslg/distro
                               /
/dev/sdd: UUID="1cbd2fb6-a8c2-49df-bb7f-0295b76b1330" BLOCK_SIZE="4096" TYPE="ext4"
/dev/sdb: BLOCK_SIZE="4096" TYPE="ext4"
/dev/sdc: UUID="56d9cb42-5f81-4936-8137-6de7508ec93d" TYPE="swap"
```

Part 3 - Mounting & Filesystem Management

```
Ajosh@DESKTOP-SSK364I:~$ sudo mkdir /mnt/devops-data
Ajosh@DESKTOP-SSK364I:~$ mount > mount-report.txt
Ajosh@DESKTOP-SSK364I:~$ df -h >> mount-report.txt
Ajosh@DESKTOP-SSK364I:~$ cat /etc/fstab >> mount-report.txt
Ajosh@DESKTOP-SSK364I:~$ cat mount-report.txt
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft
-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uuid=on)
none on /mnt/wsl type tmpfs (rw,relatime)
drivers on /usr/lib/wsl/drivers type 9p (ro,nosuid,nodev,noatime,aname=drivers;fmask=222;dmask=222,cache=0x5,access=client,msize=65536,trans=fd,rfd=8,wfd=8)
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /mnt/wslg type tmpfs (rw,relatime)
/dev/sdd on /mnt/wslg/distro type ext4 (ro,relatime,discard,errors=remount-ro,data=ordered)
none on /usr/lib/wsl/lib type overlay (rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work
,uuid=on)
rootfs on /init type rootfs (ro,size=824508k,nr_inodes=206127)
none on /dev type devtmpfs (rw,nosuid,relatime,size=824508k,nr_inodes=206127,mode=755)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000)
none on /run type tmpfs (rw,nosuid,nodev,mode=755)
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /dev/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)
none on /mnt/wslg/versions.txt type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uuid=on)
none on /mnt/wslg/doc type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uuid=on)
none on /tmp/.X11-unix type tmpfs (ro,relatime)
C:\ on /mnt/c type 9p (rw,noatime,aname=drvfs;path=C:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
D:\ on /mnt/d type 9p (rw,noatime,aname=drvfs;path=D:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
E:\ on /mnt/e type 9p (rw,noatime,aname=drvfs;path=E:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
J:\ on /mnt/j type 9p (rw,noatime,aname=drvfs;path=J:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
none on /run/user type tmpfs (rw,relatime)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,pagesize=2M)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /tmp type tmpfs (rw,nosuid,nodev,size=830268k,nr_inodes=1048576)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
none on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
```

Part 4 - Backup & Compression

```
Ajosh@DESKTOP-SSK364I:~$ tar -cvf project-backup.tar project/
project/
project/storage-report.txt
Ajosh@DESKTOP-SSK364I:~$ ls -lh project-backup.tar
-rw-r--r-- 1 Ajosh Ajosh 10K Oct 11 09:27 project-backup.tar
Ajosh@DESKTOP-SSK364I:~$ tar -xvf project-backup.tar
project/
project/storage-report.txt
Ajosh@DESKTOP-SSK364I:~$ ls -lh project-backup.tar
-rw-r--r-- 1 Ajosh Ajosh 10K Oct 11 09:27 project-backup.tar
Ajosh@DESKTOP-SSK364I:~$ tar -czvf project-backup.tar.gz project/
project/
project/storage-report.txt
Ajosh@DESKTOP-SSK364I:~$ ls -lh project-backup.tar.gz
-rw-r--r-- 1 Ajosh Ajosh 1.1K Oct 11 09:30 project-backup.tar.gz
Ajosh@DESKTOP-SSK364I:~$ tar -xzvf project-backup.tar.gz
project/
project/storage-report.txt
Ajosh@DESKTOP-SSK364I:~$ ls -lh project-backup.tar.gz
-rw-r--r-- 1 Ajosh Ajosh 1.1K Oct 11 09:30 project-backup.tar.gz
Ajosh@DESKTOP-SSK364I:~$ zip -r project.zip project/
  adding: project/ (stored 0%)
  adding: project/storage-report.txt (deflated 84%)
Ajosh@DESKTOP-SSK364I:~$ ls -lh project.zip
-rw-r--r-- 1 Ajosh Ajosh 1.3K Oct 11 09:31 project.zip
Ajosh@DESKTOP-SSK364I:~$ unzip project.zip
Archive:  project.zip
replace project/storage-report.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
  inflating: project/storage-report.txt
Ajosh@DESKTOP-SSK364I:~$ ls -lh project.zip
-rw-r--r-- 1 Ajosh Ajosh 1.3K Oct 11 09:31 project.zip
Ajosh@DESKTOP-SSK364I:~$ rsync -av project/ backup/
sending incremental file list
./
storage-report.txt

sent 5,944 bytes  received 38 bytes  11,964.00 bytes/sec
total size is 5,808  speedup is 0.97
Ajosh@DESKTOP-SSK364I:~$ ls -lh project-backup.tar project-backup.tar.gz project.zip
-rw-r--r-- 1 Ajosh Ajosh  10K Oct 11 09:27 project-backup.tar
-rw-r--r-- 1 Ajosh Ajosh 1.1K Oct 11 09:30 project-backup.tar.gz
-rw-r--r-- 1 Ajosh Ajosh 1.3K Oct 11 09:31 project.zip
Ajosh@DESKTOP-SSK364I:~$
```

Part 5 - Production Troubleshooting Challenge

1. Which command identifies disk usage?

```
Ajosh@DESKTOP-SSK364I:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            811M     0  811M   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            811M  4.0K  811M   1% /mnt/wsl
drivers         465G   77G  389G  17% /usr/lib/wsl/drivers
/dev/sdd       1007G  2.0G  954G   1% /
none            811M   52K  811M   1% /mnt/wslg
none            811M     0  811M   0% /usr/lib/wsl/lib
rootfs          806M  2.8M  803M   1% /init
none            811M  548K  811M   1% /run
none            811M     0  811M   0% /run/lock
none            811M     0  811M   0% /run/shm
none            811M   80K  811M   1% /mnt/wslg/versions.txt
none            811M   80K  811M   1% /mnt/wslg/doc
C:\             465G   77G  389G  17% /mnt/c
D:\             310G   83G  228G  27% /mnt/d
E:\             311G  375M  311G   1% /mnt/e
J:\             311G  105M  311G   1% /mnt/j
tmpfs           811M     0  811M   0% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           163M   12K  163M   1% /run/user/1000
Ajosh@DESKTOP-SSK364I:~$
```

2. Which command shows folder size?

```
Ajosh@DESKTOP-SSK364I:~$ du -sh project
12K     project
Ajosh@DESKTOP-SSK364I:~$
```

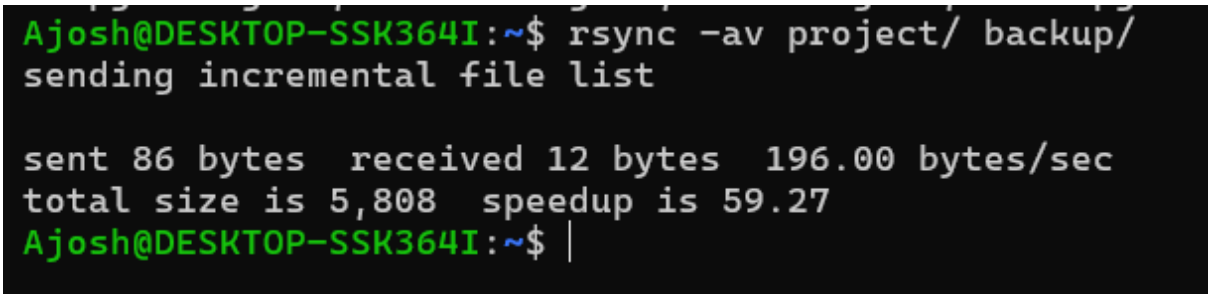
3. Which command shows available disks?

```
Ajosh@DESKTOP-SSK364I:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0     1G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /var/lib/docker
                                /mnt/wslg/distro
                                /
Ajosh@DESKTOP-SSK364I:~$
```

4. Which command removes unused packages?

```
Ajosh@DESKTOP-SSK364I:~$ sudo apt autoremove
[sudo: authenticate] Password:
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 14
Ajosh@DESKTOP-SSK364I:~$
```

5. Which command is commonly used for production backups?



1. **What is APT?**
   APT (Advanced Package Tool) is the package manager for Ubuntu/Debian. It automates software installation, updates, dependency resolution, and removal.

2. **`apt update` vs `apt upgrade`?**
   `apt update` refreshes the local package index from online repositories. `apt upgrade` upgrades the installed packages to their latest versions.

3. **What does `df -h` display?**
   Disk space usage of all mounted filesystems in human-readable units (KB, MB, GB): total size, used, available, and mount points.

4. **`df -h` vs `lsblk`?**
   `df -h` shows used and available space on currently mounted filesystems. `lsblk` shows the block devices and partitions in a tree, whether or not they are mounted.

5. **What is UUID?**
   A Universally Unique Identifier assigned to a device or filesystem. `sudo blkid` displays it along with the filesystem type and label.

6. **What is mounting in Linux?**
   Attaching a storage device or partition to a directory (mount point) in the unified filesystem tree so it can be accessed.

7. **`mount` vs `umount`?**
   `mount` attaches a storage device to a mount point. `umount` safely detaches it before removal.

8. **What is `/etc/fstab`?**
   The file system table. It is a configuration file that defines which filesystems mount automatically at system boot.

9. **`.tar` vs `.tar.gz`?**
   `.tar` bundles files into one archive without reducing size. `.tar.gz` bundles and compresses with Gzip, which cuts storage space and transfer time.

10. **Why is rsync preferred in production?**
    It efficiently synchronizes source files to a backup directory, and `-a` preserves permissions and timestamps. This makes it well suited for backups.